

The Atiyah–Hirzebruch Vanishing Theorem

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Theorem. *Let X be a compact spin manifold admitting a nontrivial S^1 -action. Then $\widehat{A}(X) = 0$.*

Proof.

Step 1. If $\dim X$ is odd, then there is nothing to prove: $\widehat{A}(X) = 0$ for degree reasons. So the argument reduces to the even-dimensional case.

Hence we may assume $\dim X = 2m$.

Step 2. Make the action isometric.

Because both X and S^1 are compact, we can average any Riemannian metric over the S^1 -action.

After replacing the metric by its average, the S^1 -action acts by isometries.

Step 3. Arrange that the action preserves the spin structure.

A spin structure on X is a 2-fold covering $P_{\text{Spin}}(X) \rightarrow P_{\text{SO}}(X)$.

If a connected Lie group G acts on X , then either G itself or a 2-fold covering group of G preserves the spin structure.

Since a 2-fold covering of S^1 is again S^1 , we may replace the given action by that lifted circle action.

Assume from now on that the S^1 -action preserves the spin structure.

Step 4. The action on spinors, equivariant index.

Because the action is by isometries and preserves the spin structure, the S^1 -action lifts to an action on the complex spinor bundle $\mathcal{S}_{\mathbb{C}} = \mathcal{S}_{\mathbb{C}}^+ \oplus \mathcal{S}_{\mathbb{C}}^-$. Since S^1 is connected, it preserves the orientation and hence the chiral splitting. This lifted action commutes with the AS operator

$$\mathcal{D}^+ : \Gamma(\mathcal{S}_{\mathbb{C}}^+) \rightarrow \Gamma(\mathcal{S}_{\mathbb{C}}^-).$$

Hence the finite-dimensional spaces $\ker \mathcal{D}^+$ and $\ker \mathcal{D}^-$ are complex S^1 -representations. Then each of them decomposes into weight spaces

$$\ker \mathcal{D}^{\pm} = \bigoplus_{k \in \mathbb{Z}} (\ker \mathcal{D}^{\pm})_k,$$

where

$$(\ker \mathcal{D}^{\pm})_k = \{v \in \ker \mathcal{D}^{\pm} \mid z \cdot v = z^k v \quad \forall z \in S^1\}.$$

Equivalently, every irreducible complex representation of S^1 is one-dimensional: $\forall n \in \mathbb{Z}$, $\exists$ a representation $\rho_n : S^1 \rightarrow \text{GL}(1, \mathbb{C})$, $e^{i\theta} \mapsto e^{in\theta}$.

So any finite-dimensional S^1 -representation V decomposes as $V \cong \bigoplus_k \rho_{n_k}$.

In particular, $\ker \mathcal{D}^+ = \bigoplus_k V_k^+$, where $V_k^+ = \{v \in \ker \mathcal{D}^+ : z \cdot v = z^k v \quad \forall z \in S^1\}$.

Similarly, $\ker \mathcal{D}^- = \bigoplus_k V_k^-$, where $V_k^- = \{v \in \ker \mathcal{D}^- : z \cdot v = z^k v \quad \forall z \in S^1\}$.

Now define the equivariant index. Since $\mathcal{D}^+$ is S^1 -equivariant, $\ker \mathcal{D}^+$ and $\ker \mathcal{D}^-$ are both finite-dimensional S^1 -representations, we may form the equivariant index:

$$\text{Ind}_{S^1}[\mathcal{D}^+] = [\ker \mathcal{D}^+] - [\ker \mathcal{D}^-] \in R(S^1).$$

Under the character isomorphism $R(S^1) \cong \mathbb{Z}[t, t^{-1}]$, with $[\rho_n] \mapsto t^n$, $t = e^{i\theta}$, and representation $\mapsto$ trace, the equivariant index becomes the Laurent polynomial $i_{\mathcal{D}^+}(t) = \sum_k (\dim V_k^+ - \dim V_k^-) t^k$.

Note $\forall g \in S^1$, $\text{tr}(g | \ker \mathcal{D}^+) = \sum_k (\dim V_k^+) g^k$, because on a space where g acts by the scalar g^k , the trace is just $g^k \cdot \text{dimension}$.

Similarly, $\text{tr}(g | \ker \mathcal{D}^-) = \sum_k (\dim V_k^-) g^k$. Then

$$\text{tr}(g | \ker \mathcal{D}^+) - \text{tr}(g | \ker \mathcal{D}^-) = \sum_k (\dim V_k^+ - \dim V_k^-) g^k = i_{\mathcal{D}^+}(g).$$

If we are evaluating at $g = 1$, we recover $i_{\mathcal{D}^+}(1) = \dim \ker \mathcal{D}^+ - \dim \ker \mathcal{D}^- = \widehat{A}(X)$.

So if we can show $i_{\mathcal{D}^+} = 0$ as a Laurent polynomial, then $\widehat{A}(X) = 0$.

Step 5. Choose a non-torsion element and identify the fixed point set.

Choose $g \in S^1$ which is not a root of unity. Then the subgroup generated by g is dense in S^1 : $\langle g \rangle = \{g^n : n \in \mathbb{Z}\}$.

Let

$$\bar{F}_g = X^g = \{x \in X : g \cdot x = x\}, \quad \bar{F} = X^{S^1} = \{x \in X : z \cdot x = x \quad \forall z \in S^1\}.$$

We claim that $\bar{F}_g = \bar{F}$.

$\bar{F} \subseteq \bar{F}_g$: if $x \in \bar{F} = X^{S^1}$, then $z \cdot x = x \quad \forall z \in S^1$. Since $g \in S^1$, $g \cdot x = x$, so $x \in \bar{F}_g$.

$\bar{F}_g \subseteq \bar{F}$: if $x \in \bar{F}_g$, then $g^n \cdot x = x \quad \forall n \in \mathbb{Z}$.

Since $\langle g \rangle$ is dense in S^1 , for any $h \in S^1$, $\exists$ a sequence $n_1, \dots, n_j$ such that $g^{n_j} \rightarrow h$ in S^1 . By continuity of the action,

$$h \cdot x = \lim_{j \rightarrow \infty} g^{n_j} \cdot x = \lim_{j \rightarrow \infty} x = x.$$

h was chosen arbitrarily, so $x \in \bar{F}$, and therefore $\bar{F}_g = \bar{F}$.

Step 6. G-spin theorem.

Let N be the normal bundle of $\bar{F}$ in X . Because $\bar{F}$ is fixed by S^1 , the S^1 -action preserves N . N decomposes S^1 -equivariantly as $N = \bigoplus_{m \geq 0} N_m$, where each N_m is a complex vector bundle. When $g \in S^1$ acts on $v \in N_m$, $g \cdot v = g^m \cdot v$, where $g^m \in \mathbb{C}$ ($g \in S^1 \subset \mathbb{C}$).

We want to compute $i_{p^+}(g)$ for non-torsion g .

For a complex rank k bundle E with formal Chern roots $x_1, \dots, x_k$, define the characteristic function

$$V(E, t) = \prod_{j=1}^k \left(t^{1/2} e^{x_j/2} - t^{-1/2} e^{-x_j/2} \right)^{-1}.$$

Because

$$\widehat{A}_\theta(E) = \prod_{j=1}^k \left(\underbrace{e^{\frac{1}{2}(x_j + i\theta)}}_{:= \textcircled{1}} - \underbrace{e^{-\frac{1}{2}(x_j + i\theta)}}_{:= \textcircled{2}} \right)^{-1},$$

if we set $t = e^{i\theta}$, then $V(E, t)$ is the characteristic class $\widehat{A}_\theta(E)$ rewritten in $t = e^{i\theta}$.

Because $\bar{F}_g = \bar{F}$, the formula becomes, componentwise on $\bar{F}$,

$$\text{Spin}(X, g) = (-1)^\delta \left\{ \prod_{0 < \theta \leq \pi} \widehat{A}_\theta(N(\theta)) \cdot \widehat{A}(\bar{F}) \right\} [\bar{F}],$$

where $(-1)^\delta$ is locally constant on the components of $\bar{F}$.

In the G-spin theorem, the normal bundle decomposes by rotation angle: $N = \bigoplus N(\theta)$, and dg rotates vectors in $N(\theta)$ by angle θ .

In our case, $g = e^{i\theta_0} \in S^1$ acts on N_m by $g^m = e^{im\theta_0}$. So the factor coming from N_m is rewritten by setting $e^{i\theta} = g^m$.

From $\widehat{A}_\theta(E) = V(E, e^{i\theta})$, we compute the corresponding factor as

$$V(N_m, e^{im\theta_0}) = V(N_m, g^m).$$

By definition,

$$\text{Spin}(X, g) = \text{ind}_g(\mathcal{D}^+) = i_{p^+}(g) = (-1)^\delta \left\{ \prod_m V(N_m, g^m) \widehat{A}(\bar{F}) \right\} [\bar{F}]. \quad (*)$$

LHS: $i_{p^+}(g)$ is the Laurent polynomial $\sum_k a_k t^k$ evaluated at g , in particular a rational function.

RHS: Look at $V(N_m, g^m)$. Each factor is $(g^{m/2} e^{x_j/2} - g^{-m/2} e^{-x_j/2})^{-1}$. Although $V(N_m, g^m)$ is defined only up to sign because of the square roots, the G-spin theorem fixes the signs on the components of $\bar{F}$. After multiplying by $\widehat{A}(\bar{F})$ and evaluating on $[\bar{F}]$, the whole RHS gives the rational function in g which agrees with $i_{p^+}(g)$.

Both sides of $(*)$ are rational functions of g . This identity holds for all non-torsion g . We replace g by a variable t .

Step 7. WTS: RHS vanishes as $t \rightarrow 0$ and as $t \rightarrow \infty$.

Case ①: If $\bar{F} = \emptyset$, $\text{RHS} = 0$ for all non-torsion g , so the Laurent polynomial $i_{p^+}(t)$ is identically zero, and we are done.

Case ②: Suppose $\bar{F} \neq \emptyset$. Since the S^1 -action is nontrivial, $\bar{F} \neq X$; otherwise every point would be fixed and the action would be trivial.

So $\bar{F}$ is a proper submanifold of X , $\dim \bar{F} < \dim X$, and the normal bundle N is nonzero. Since $N = \bigoplus_{m>0} N_m$ as a real bundle with complex summands and $\text{rank}_{\mathbb{R}}(N) = 2 \sum_{m>0} \text{rank}_{\mathbb{C}}(N_m) > 0$, at least one N_m has positive complex rank.

Now we are looking at $V(E, t)$.

$$\begin{aligned} V(E, t) &= \prod_{j=1}^k \left(t^{1/2} e^{x_j/2} - t^{-1/2} e^{-x_j/2} \right)^{-1} \\ &= \prod_{j=1}^k t^{1/2} e^{x_j/2} (t e^{x_j} - 1)^{-1} \quad \textcircled{1} \\ &= \prod_{j=1}^k t^{-1/2} e^{-x_j/2} (1 - t^{-1} e^{-x_j})^{-1} \quad \textcircled{2}. \end{aligned}$$

$$\left. \begin{array}{l} \textcircled{1} : \text{ if } t \rightarrow 0, \quad t^{1/2} e^{x_j/2} \rightarrow 0, \\ \textcircled{2} : \text{ if } t \rightarrow \infty, \quad t^{-1/2} e^{-x_j/2} \rightarrow 0, \end{array} \right\} \quad \text{whenever } \text{rank}_{\mathbb{C}}(E) > 0.$$

Thus $V(E, t) \rightarrow 0$ as $t \rightarrow 0$ and as $t \rightarrow \infty$. We apply this to at least one positive complex-rank summand N_m , so $\prod_m V(N_m, t^m)$ vanishes as $t \rightarrow 0$ and as $t \rightarrow \infty$.

So the RHS vanishes at both limits: $i_{p+}(t) \rightarrow 0$ as $t \rightarrow 0$ and as $t \rightarrow \infty$.

If $i_{p+}(t)$ is not identically zero, write $i_{p+}(t) = \sum_{j=p}^q a_j t^j$ with $a_p \neq 0$ and $a_q \neq 0$.

$$\begin{array}{ll} \text{If } p < 0, & a_p t^p \rightarrow \infty \text{ as } \boxed{t \rightarrow 0}, \text{ but } i_{p+}(t) \rightarrow 0, \\ p = 0, & i_{p+}(t) \rightarrow a_0 \neq 0, \end{array} \quad \implies \quad p > 0.$$

$$\begin{array}{ll} \text{If } q > 0, & a_q t^q \rightarrow \infty \text{ as } \boxed{t \rightarrow \infty}, \text{ but } i_{p+}(t) \rightarrow 0, \\ q = 0, & i_{p+}(t) \rightarrow a_0 \neq 0, \end{array} \quad \implies \quad q < 0.$$

But $p \leq q$ always. Therefore $i_{p+}(t) \equiv 0$.

Finally, $0 = i_{p+}(1) = \dim \ker \not{D}^+ - \dim \ker \not{D}^- = \hat{A}(X)$.

□

References

- [1] H. Blaine Lawson, Jr. and Marie-Louise Michelsohn, *Spin Geometry*, Princeton Mathematical Series, vol. 38, Princeton University Press, Princeton, NJ, 1989.